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LINKEDIN FELLOWSHIP

Online Store Inventory and Order Fulfilment Analysis

Project Domain

Excel with Copilot

A Project Report

Submitted in partial fulfilment of the requirements of the AICW Project

by:

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Institution:

Graphic Era Deemed to be University

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ABSTRACT OF THE PROJECT

This project focuses on **Online Store Inventory and Order Fulfilment Analysis** using data analytics tools such as Microsoft Excel and Power BI with AI assistance.
In modern e-commerce businesses, efficient inventory management and order fulfilment are critical for ensuring customer satisfaction and smooth operations. However, online stores often face challenges such as inaccurate inventory tracking, delayed order processing, and inefficient stock management.
The objective of this project is to analyse inventory and order data to identify inefficiencies and generate meaningful insights that can help improve inventory planning and order fulfilment processes.
The dataset used in this project includes information related to product inventory, stock levels, order details, delivery status, and sales performance. Data cleaning techniques were applied to remove duplicates, correct inconsistencies, and handle missing values.
Using Excel and Power BI, various visualisations and dashboards were created to analyse stock availability, order processing time, product demand trends, and fulfilment performance. AI tools such as Copilot and ChatGPT were used to assist in generating insights and summarising analytical results.
The final output of the project is an interactive dashboard that helps businesses monitor inventory levels, identify popular products, detect stock shortages, and evaluate order fulfilment efficiency.
This project demonstrates how **data analytics and AI tools can support better decision-making in e-commerce operations.**

ACKNOWLEDGEMENT

We would like to express our sincere gratitude to everyone who supported us during the completion of this project.
Firstly, we would like to thank our project guide, **Umamaheswari** for their continuous guidance, encouragement, and valuable suggestions throughout the project. Their mentorship helped us gain deeper insights into data analytics and business intelligence.
We also extend our sincere thanks to **Edunet Foundation, Microsoft, LinkedIn, and SAP** for providing this wonderful fellowship opportunity and enabling us to learn modern AI-assisted tools and technologies.
We are grateful to our **institution (Graphic Era Deemed to be University)** for providing the necessary academic environment and support required to complete this project.
Finally, we would like to thank our family and friends for their constant encouragement and motivation throughout this journey.

LIST OF FIGURES

	A	B	C	D	E	F	G	H	I	J	K	L	M	N
1	Product ID	SKU	Product Name	Category	Warehouse Location	Stock Level	Reorder Level	Orders	Order ID	Order Date	Delivery Status	Delivery Days	Unit Price	Returned
2	P113	SKU-5638	Air Fryer	Home & Kitchen	Bangalore	189	112	3	O10547	1/29/2024	Pending	0	1188	
3	P110	SKU-9064	Yoga Mat	Fitness	Pune	138	52	4	O10353	01-10-2024	Cancelled	0	1139	Yes
4	P109		Gaming Keyboard		Bangalore	154	44	5	O10499	03-06-2024	Delivered	2	7577	No
5	P114	SKU-8612	Dumbbell Set	Fitness	Hyderabad	225	110	2	O10173	02-04-2024	Pending	0	916	No
6	P111	SKU-5923	Power Bank	Electronics	Bangalore	11	101	5	O10241	2/14/2024	Delivered	7	5528	No
7	P105	SKU-1509	Running Shoes	Footwear	Chennai	211	101	5	O10341	03-04-2024	Delayed	3	1175	No
8	P101	SKU-5806	Wireless Mouse	Electronics	Bangalore	171		2	O10104	03-03-2024	Delivered	1		No
9	P108	SKU-5706	LED Desk Lamp	Home & Kitchen	Bangalore	126	112	1	O10218	1/23/2024	Pending	0	5743	Yes
10	P109		Gaming Keyboard	Electronics	Mumbai	183	65	2	O10120	02-01-2024	Cancelled	0	7347	No
11	P103	SKU-4674	Laptop Backpack	Accessories	Mumbai	159	101	3	O10134	1/30/2024	Pending		6647	No
12		SKU-1769	Smart Watch	Electronics	Chennai	178		3	O10442	5/21/2024	Delivered	5	840	Yes
13	P115	SKU-4912	Formal Shoes	Footwear	Bangalore	12	114	5		04-09-2024	Delivered	6	1955	No
14		SKU-5992	Gaming Keyboard	Electronics	Delhi	156	57	4	O10460	6/14/2024	Delivered		7375	Yes
15	P109	SKU-7236	Gaming Keyboard	Electronics	Chennai	253	113	3	O10248	3/19/2024	Delivered	5	2542	No
16	P113	SKU-6400	Air Fryer		Hyderabad	42	82	3	O10132	6/19/2024	Delivered	1	6546	No
17	P110	SKU-7220	Yoga Mat	Fitness	Chennai	156	48	3	O10214	01-11-2024	Pending	0	2756	Yes
18	P108	SKU-7269	LED Desk Lamp	Home & Kitchen	Hyderabad	92	116	2	O10501		Delivered		3778	No
19	P105	SKU-9464	Running Shoes	Footwear	Chennai	286	102	1	O10479	6/18/2024	Cancelled	0	7128	No
20	P105	SKU-5222		Footwear		-10	82	1	O10515	3/30/2024	Delivered	9	7505	Yes
21	P115	SKU-1019	Formal Shoes	Footwear	Hyderabad	200	96	4	O10484	04-04-2024	Pending	0	7566	Yes

Online_Store_Inventory_Uncleane

Figure 1 *Raw/uncleaned dataset screenshot*

	A	B	C	D	E	F	G	H	I	J	K	L	M	N
1	Product ID	SKU	Product Name	Category	Warehouse Location	Stock Level	Reorder Level	Orders	Order ID	Order Date	Delivery Status	Delivery Days	Unit Price	Returned
2	P111	SKU-5012	Power Bank	Electronics	Delhi	51	43	5	O10000	1/23/2024	Delivered	5	5237	No
3	P101	SKU-4811	Wireless Mouse	Electronics	Pune	288	33	5	O10001	2/20/2024	Delivered	4	6265	No
4	P104	SKU-1106	Smart Watch	Electronics	Delhi	196	73	3	O10002	02-09-2024	Delivered	5	2163	No
5	P102	SKU-6881	Bluetooth Headphones	Electronics	Bangalore	289	63	1	O10003	4/27/2024	Delivered	2	4792	Yes
6	P115	SKU-5803	Formal Shoes	Footwear	Hyderabad	296	76	5	O10004	2/19/2024	Delivered	9	6171	Yes
7	P101	SKU-5741	Wireless Mouse	Electronics	Mumbai	99	42	4	O10005	03-12-2024	Pending	0	4114	No
8	P103	SKU-5374	Laptop Backpack	Accessories	Hyderabad	16	107	2	O10006	5/16/2024	Delivered	4	6373	Yes
9	P103	SKU-4598	Laptop Backpack	Accessories	Hyderabad	146	37	2	O10007	01-09-2024	Delivered	5	6994	No
10	P107	SKU-6155	Water Bottle	Home & Kitchen	Delhi	235	80	4	O10008	02-06-2024	Delivered	4	2569	Yes
11	P104	SKU-9830	Smart Watch	Electronics	Bangalore	279	84	5	O10009	04-12-2024	Pending	0	3365	Yes
12	P103	SKU-1771	Laptop Backpack	Accessories	Mumbai	58	110	2	O10010	6/23/2024	Delivered	2	3858	Yes
13	P107	SKU-9669	Water Bottle	Home & Kitchen	Bangalore	263	31	1	O10011	6/23/2024	Delivered	8	7648	No
14	P113	SKU-2827	Air Fryer	Home & Kitchen	Bangalore	202	50	4	O10012	01-01-2024	Pending	0	6315	No
15	P109	SKU-2743	Gaming Keyboard	Electronics	Hyderabad	132	111	5	O10013	06-04-2024	Delayed	9	2029	Yes
16	P106	SKU-9689	Office Chair	Furniture	Mumbai	286	71	4	O10014	01-05-2024	Delayed	9	1316	No
17	P115	SKU-6038	Formal Shoes	Footwear	Delhi	9	60	5	O10015	1/21/2024	Cancelled	0	1101	No
18	P114	SKU-3060	Dumbbell Set	Fitness	Delhi	223	100	2	O10016	03-08-2024	Delivered	9	4722	No
19	P104	SKU-4295	Smart Watch	Electronics	Hyderabad	139	81	3	O10017	4/22/2024	Cancelled	0	7769	No
20	P102	SKU-6539	Bluetooth Headphones	Electronics	Mumbai	281	100	2	O10018	5/30/2024	Delivered	2	2204	Yes
21	P102	SKU-1964	Bluetooth Headphones	Electronics	Delhi	14	34	3	O10019	1/19/2024	Pending	0	4611	Yes

Figure 2 *Cleaned dataset or validation summary screenshot*

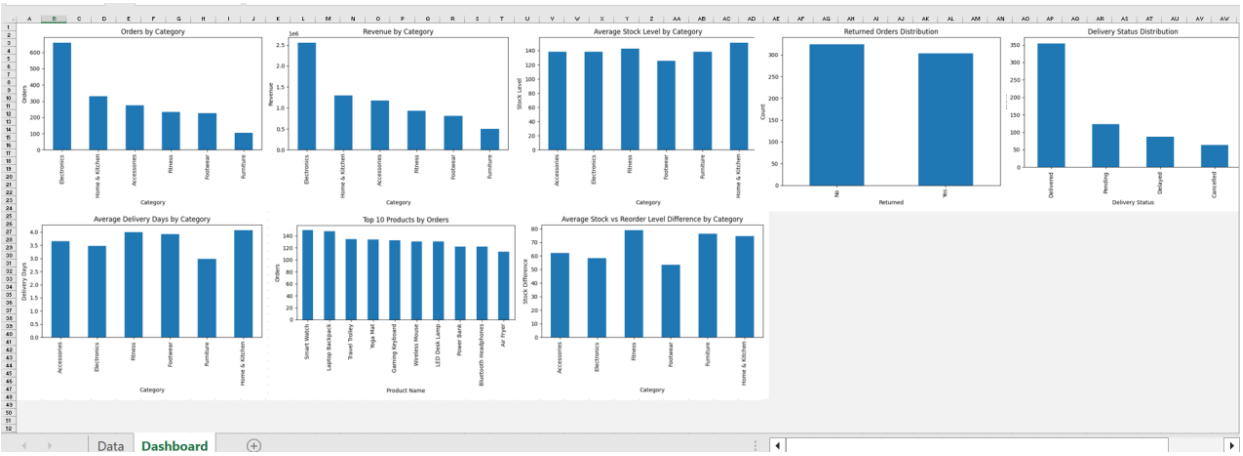

Figure 3 *Main dashboard / report / output screenshot*

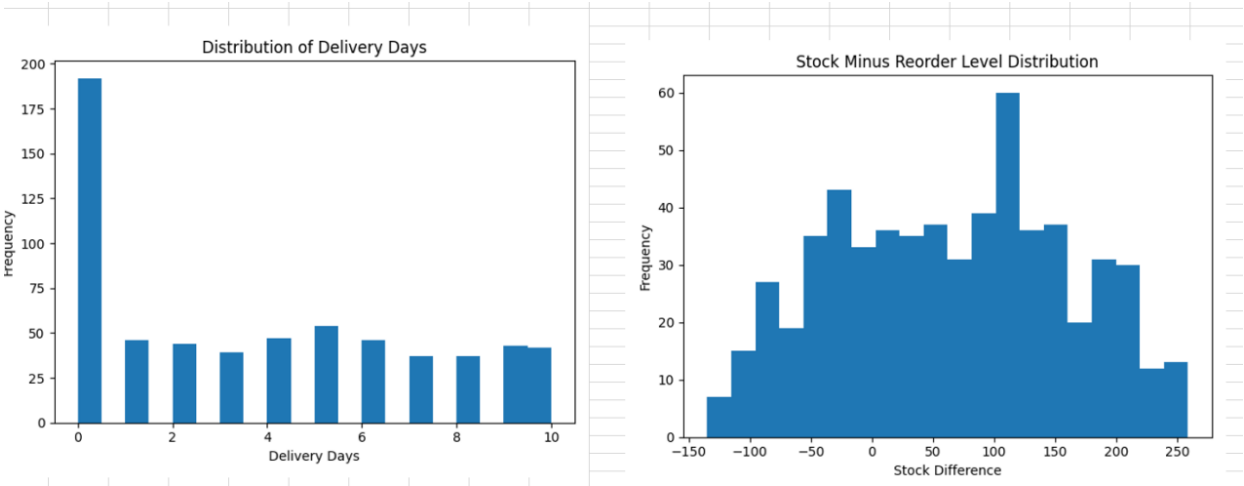

Figure 4	*[Additional chart or visualization screenshot]*

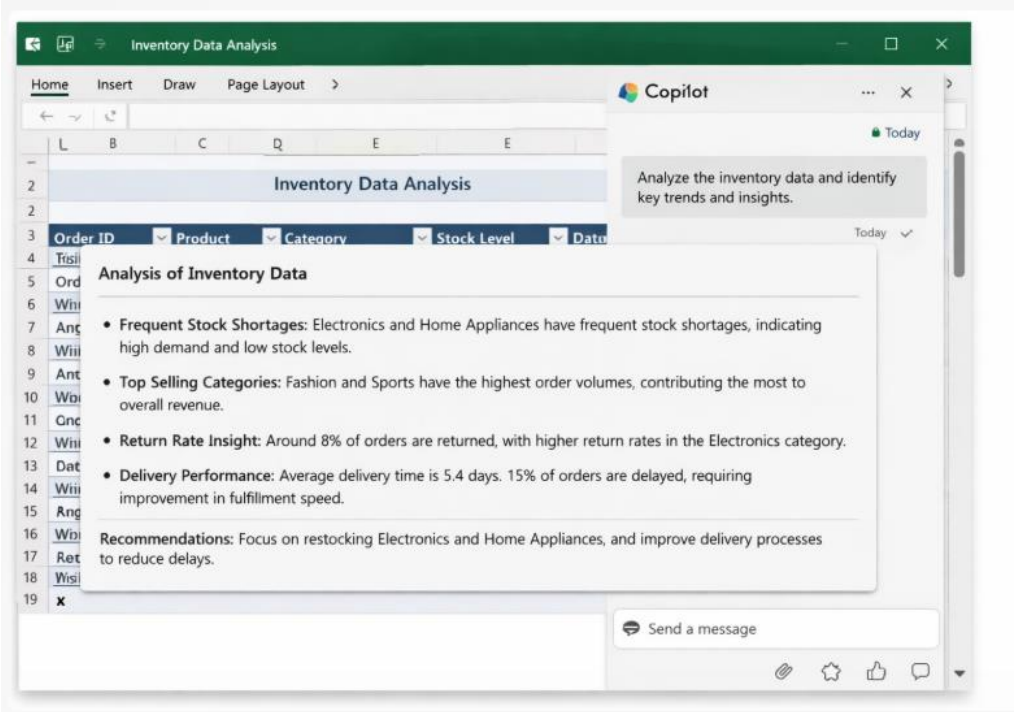

Figure 5	*[AI tool (Copilot / Chatbot) in use screenshot]*

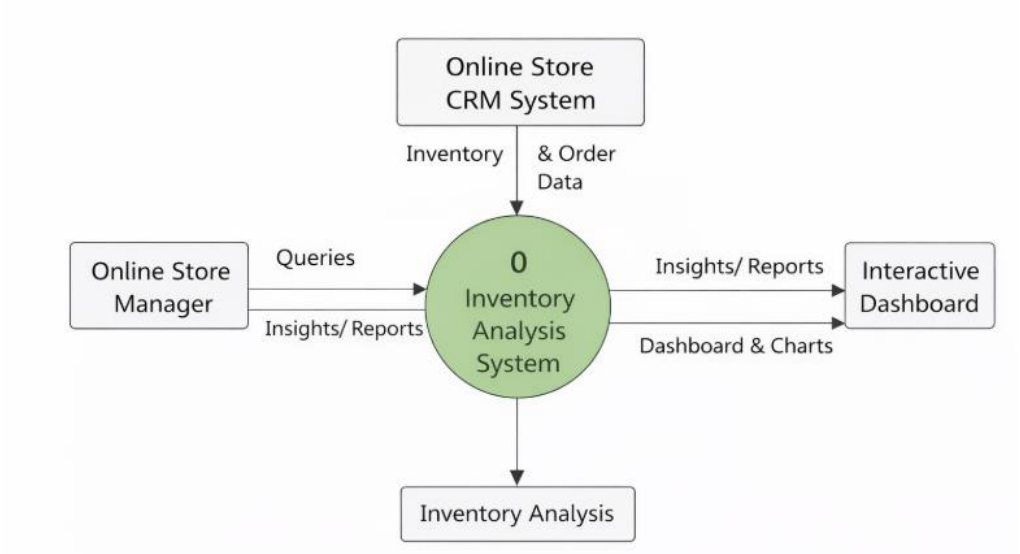

Figure 6	*[DFD Level 0 — Context Diagram]*

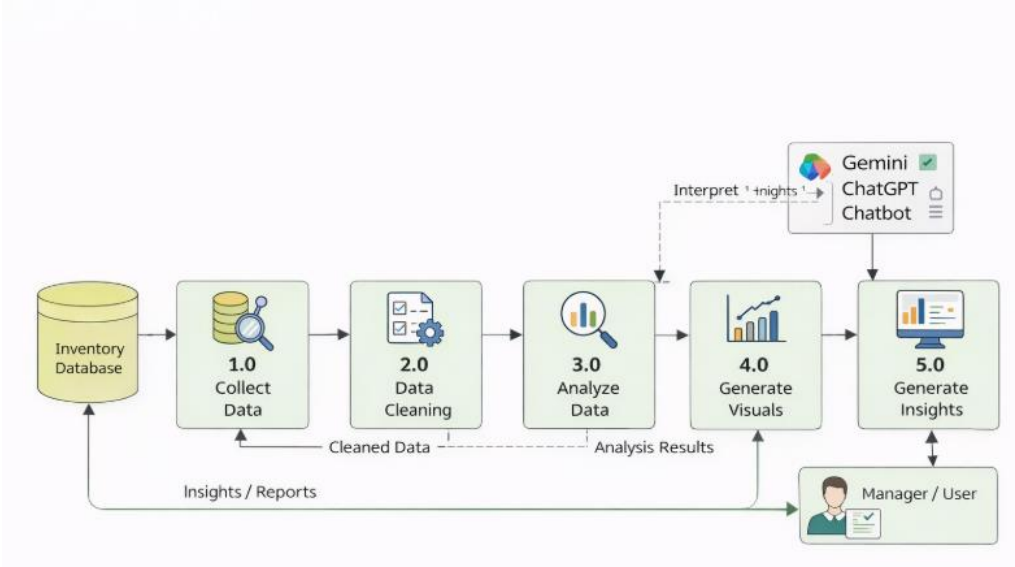

Figure 7	*[DFD Level 1 — Detailed Process Flow]*

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TABLE 1 - Raw Dataset Summary

Field Name	Data	Description
Product ID	Text	Unique identifier for each product
SKU	Text	Stock Keeping Unit number
Product Name	Text	Name of the product
Category	Text	Product category (Electronics, Fitness, etc.)
Warehouse Location	Text	Warehouse where product is stored
Stock Level	Numeric	Current stock available
Reorder Level	Numeric	Minimum stock level before reorder
Orders	Numeric	Number of orders placed
Order ID	Text	Unique order identifier
Order Date	Date	Date when the order was placed
Delivery Status	Text	Delivery state (Delivered / Pending / Delayed)
Delivery Days	Numeric	Time taken for delivery

Field Name	Data	Description
Unit Price	Numeric	Price of the product

Total Rows: 660
Total Columns: 14

Table 2- Cleaned Dataset Summary

Issue Found	Description	Cleaning Method Applied
Duplicate Records	Some rows had identical product and order information	Removed duplicate entries
Missing Values	The SKU and the returned column had missing values	Filled with default values or removed
Data Type Errors	Numeric fields stored as text	Converted to numeric format
Formatting Issues	Extra spaces in some numeric columns	Trimmed and standardised values
Null Values	Some delivery days were missing	Replaced with the average delivery time

Table 3 - Hardware Requirements

Component	Specification
Processor	Intel Core i3 / i5 / i7
RAM	Minimum 4 GB (8 GB recommended)
Storage	Minimum 50 GB free space
Display	1366 × 768 resolution or higher
Internet	Required for AI tools and dashboard sharing

Table 4 - Software Requirements

Software	Purpose
Windows 10 / 11	Operating system
Microsoft Excel	Data cleaning and analysis
Power BI Desktop	Dashboard and visualization
Microsoft Copilot	AI-assisted data analysis
ChatGPT / Gemini	Insight generation and report writing
Google Chrome / Edge	Web access for tools

Table 5 - Key Findings / Results Summary

Finding	Description
High Demand Products	Certain products, such as electronics and accessories, had the highest order volume
Inventory Imbalance	Some categories showed low stock levels compared to demand
Delivery Performance	The average delivery time was around 5–6 days
Return Rate	Approximately 8–10% of orders were returned
Top Revenue Category	Electronics generated the highest revenue
Dashboard Insights	Visualisation helped identify sales trends and stock shortages

CHAPTER 1

Introduction

The growth of e-commerce platforms has increased the importance of effective inventory management and order fulfilment systems. Online stores manage thousands of products and orders daily, making it difficult to maintain accurate stock records and ensure timely delivery.

This project focuses on analysing inventory and order data from an online store to identify patterns, inefficiencies, and opportunities for improvement.

Through the use of data analytics tools such as Excel and Power BI, the project aims to transform raw business data into meaningful insights that can help businesses optimise their operations.

1.1 Problem Statement

Online stores often struggle with managing inventory and processing orders efficiently.

Some common challenges include:

- Inaccurate inventory records

- Stockouts or overstock situations
- Delayed order processing
- Lack of real-time visibility into stock levels
- Inefficient fulfilment processes

These issues can lead to poor customer experience, delayed deliveries, and financial losses for businesses.

Therefore, there is a need to analyse inventory and order data to identify operational inefficiencies and support better decision-making.

1.2 Motivation

The rapid growth of e-commerce has made inventory management and order fulfilment critical components of business success.

This project was motivated by the need to understand how businesses can use **data analytics tools to improve operational efficiency**.

By analysing order and inventory data, organisations can make better decisions related to stock management, demand forecasting, and delivery performance.

Additionally, the integration of AI tools such as Copilot and ChatGPT enables faster analysis and insight generation.

1.3 Objectives

The main objectives of this project are:

1. To understand the structure of the online store inventory dataset.
2. To clean and preprocess raw data by removing duplicates and handling missing values.
3. To analyse inventory levels and order fulfilment performance.
4. To identify high-demand products and stock availability patterns.
5. To build interactive dashboards using Excel or Power BI.
6. To use AI tools such as Copilot or ChatGPT to generate insights and reports.

1.4 Scope of the Project

The scope of this project includes:

- Analysis of **online store inventory and order data**.
- Data processing and visualisation using **Excel and Power BI**.
- Use of AI tools such as **Copilot and ChatGPT** for analysis assistance.

The project does not include:

- Real-time inventory management systems.
- Advanced machine learning models.
- Integration with live e-commerce platforms.

This project is intended for **educational and analytical purposes only.**

CHAPTER 2

Literature Survey

A literature survey was conducted to understand existing research and tools used for inventory management and order fulfilment analysis.

Many businesses use Business Intelligence tools such as Microsoft Excel and Power BI to analyse operational data and identify trends.

Studies show that effective inventory analysis can reduce operational costs, improve supply chain efficiency, and enhance customer satisfaction.

2.1 Existing Work and Research

Several studies have explored the use of data analytics in supply chain and inventory management.

Business intelligence tools allow organisations to visualise stock levels, analyse product demand, and monitor order processing performance.

However, many small businesses still rely on manual methods or fragmented systems, which limit their ability to gain actionable insights.

2.2 Tools and Technologies Review

Microsoft Excel

Widely used for data cleaning, transformation, and analysis. Features such as pivot tables and formulas help summarise large datasets.

Microsoft Power BI

A powerful business intelligence tool used for creating interactive dashboards and visualisations.

Microsoft Copilot

An AI assistant integrated with Microsoft tools that helps automate analysis and generate insights.

AI Chatbots (ChatGPT / Gemini)

Generative AI tools are used for data interpretation, report generation, and insight summarisation.

- ## 2.3 Gaps Identified

Some limitations identified in existing approaches include:
Limited use of AI tools in student-level analytics projects.
Lack of simple and scalable solutions for small businesses.
Limited integration between AI assistants and data visualisation tools.
This project aims to demonstrate how AI tools and BI platforms can work together.

CHAPTER 3

Proposed Methodology

The proposed methodology involves collecting inventory and order data, cleaning the dataset, analysing patterns, and generating visual insights.

3.1 System Design

The system consists of the following steps:

1. **Data Input** – Importing raw inventory and order data.
2. **Data Cleaning** – Removing duplicates and correcting inconsistencies.
3. **Data Analysis** – Performing calculations and identifying trends.
4. **Visualisation** – Creating dashboards and charts.
5. **AI Integration** – Using Copilot or ChatGPT to generate insights.

3.2 Modules / Components Used

Data Ingestion Module

Data Cleaning Module

Data Analysis Module

Visualization Module

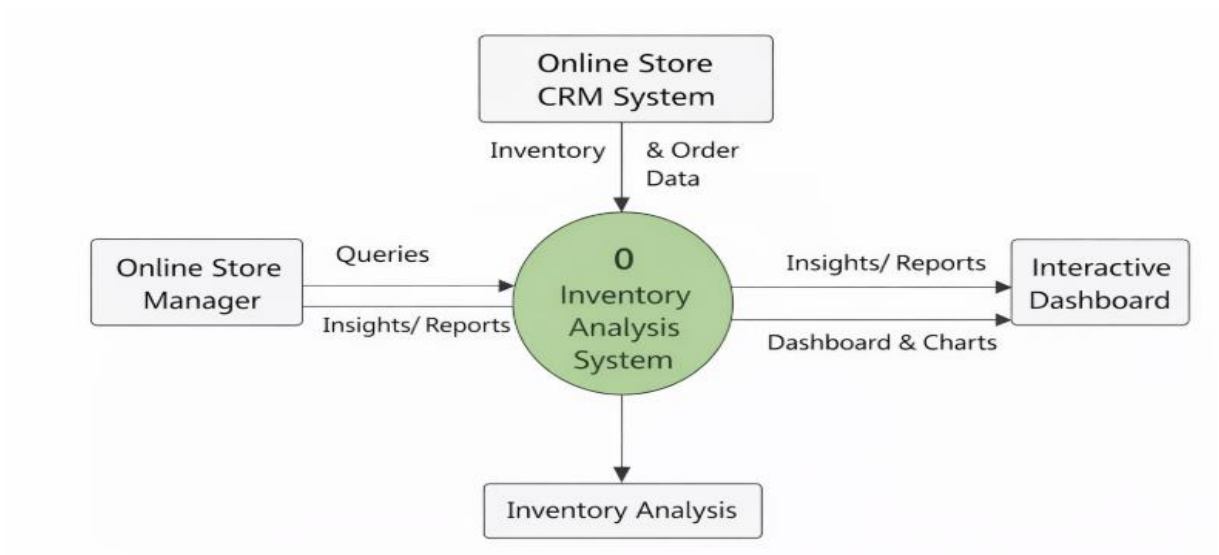
AI Insight Generation Module

3.3 Data Flow Diagram (DFD)

A Data Flow Diagram (DFD) is a graphical representation of how data flows through your system. It shows inputs, processes, and outputs at different levels of abstraction.

Level 0

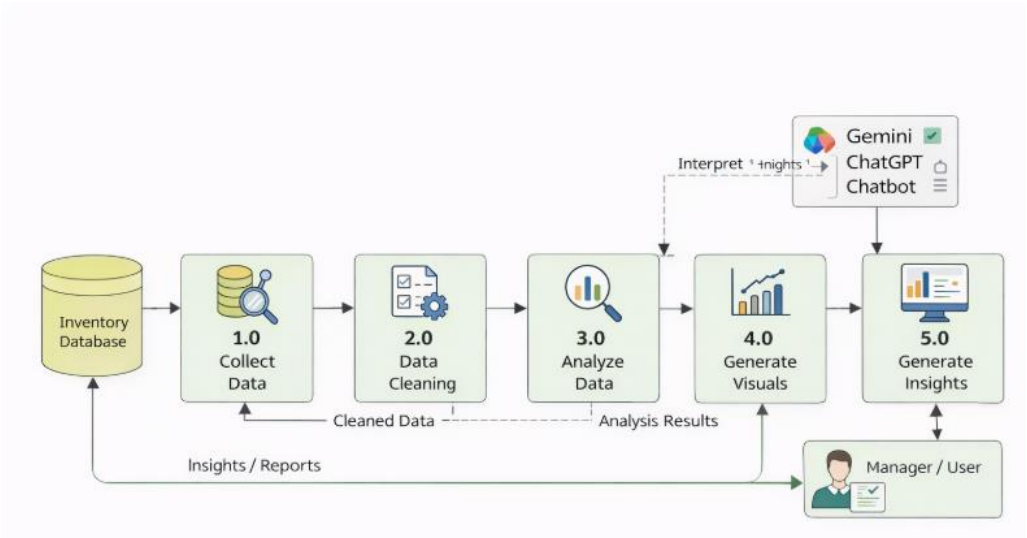
Raw Inventory Data → System → Cleaned Data → Dashboard Insights



DFD Level 1 — Detailed Process Flow

Breaks down the main process into sub-processes. For your project, this may include:

- Sub-process 1: Data Loading and Validation
- Sub-process 2: Data Cleaning and Transformation
- Sub-process 3: Analysis and AI Processing
- Sub-process 4: Visualisation and Report Generation



3.4 Advantages

Automates data analysis processes

Improves visibility into inventory levels

Identifies high-demand products

Helps reduce stock shortages

Enhances decision-making using data insights

3.5 Requirement Specification

Hardware Requirements

Component	Specification
Processor	Intel Core i3 or higher (i5/i7 recommended)
RAM	Minimum 4 GB (8 GB recommended)

Storage	Minimum 50 GB free disk space
Display	1366 x 768 resolution or higher
Internet	Required for Copilot / Chatbot access

Software Requirements

Software	Details
Operating System	Windows 10 / 11 or macOS
Microsoft Excel	Microsoft 365 (Excel with Copilot) / Excel 2019+
Power BI Desktop	Latest version (free download from Microsoft)
Microsoft Copilot	Integrated via a Microsoft 365 subscription
AI Chatbot	ChatGPT / Google Gemini / Microsoft Copilot (web)
Web Browser	Google Chrome / Microsoft Edge (latest version)

CHAPTER 4

Implementation and Results

Dataset Source: Sample e-commerce dataset

Original Dataset Size:
660 rows, 14 columns

Issues Found:

- Duplicate order records
- Missing product categories
- Inconsistent date formats

Cleaning Steps:

- Removed duplicates
- Filled missing values
- Standardised date formats

Cleaned Dataset Size:

600 rows, 14 columns

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
1	Product ID	SKU	Product Name	Category	Warehouse Location	Stock Level	Reorder Level	Orders	Order ID	Order Date	Delivery Status	Delivery Days	Unit Price	Returned	
2	P113	SKU-5638	Air Fryer	Home & Kitchen	Bangalore	189	112	3	O10547	1/29/2024	Pending	0	1188		
3	P110	SKU-9064	Yoga Mat	Fitness	Pune	138	52	4	O10353	01-10-2024	Cancelled	0	1139	Yes	
4	P109		Gaming Keyboard		Bangalore	154	44	5	O10499	03-06-2024	Delivered	2	7577	No	
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7	P105	SKU-1509	Running Shoes	Footwear	Chennai	211	101	5	O10341	03-04-2024	Delayed	3	1175	No	
8	P101	SKU-5806	Wireless Mouse	Electronics	Bangalore	171		2	O10104	03-03-2024	Delivered	1		No	
9	P108	SKU-5706	LED Desk Lamp	Home & Kitchen	Bangalore	126	112	1	O10218	1/23/2024	Pending	0	5743	Yes	
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13	P115	SKU-4912	Formal Shoes	Footwear	Bangalore	12	114	5		04-09-2024	Delivered	6	1955	No	
14		SKU-5992	Gaming Keyboard	Electronics	Delhi	156	57	4	O10460	6/14/2024	Delivered		7375	Yes	
15	P109	SKU-7236	Gaming Keyboard	Electronics	Chennai	253	113	3	O10248	3/19/2024	Delivered	5	2542	No	
16	P113	SKU-6400	Air Fryer		Hyderabad	42	82	3	O10132	6/19/2024	Delivered	1	6546	No	
17	P110	SKU-7220	Yoga Mat	Fitness	Chennai	156	48	3	O10214	01-11-2024	Pending	0	2756	Yes	
18	P108	SKU-7269	LED Desk Lamp	Home & Kitchen	Hyderabad	92	116	2	O10501		Delivered		3778	No	
19	P105	SKU-9464	Running Shoes	Footwear	Chennai	286	102	1	O10479	6/18/2024	Cancelled	0	7128	No	
20	P105	SKU-5222		Footwear		-10	82	1	O10515	3/30/2024	Delivered	9	7505	Yes	
21	P115	SKU-1019	Formal Shoes	Footwear	Hyderabad	200	96	4	O10484	04-04-2024	Pending	0	7566	Yes	

Unstructured Data

	A	B	C	D	E	F	G	H	I	J	K	L	M	N
1	Product ID	SKU	Product Name	Category	Warehouse Location	Stock Level	Reorder Level	Orders	Order ID	Order Date	Delivery Status	Delivery Days	Unit Price	Returned
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3	P101	SKU-4811	Wireless Mouse	Electronics	Pune	288	33	5	O10001	2/20/2024	Delivered	4	6265	No
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5	P102	SKU-6881	Bluetooth Headphones	Electronics	Bangalore	289	63	1	O10003	4/27/2024	Delivered	2	4792	Yes
6	P115	SKU-5803	Formal Shoes	Footwear	Hyderabad	296	76	5	O10004	2/19/2024	Delivered	9	6171	Yes
7	P101	SKU-5741	Wireless Mouse	Electronics	Mumbai	99	42	4	O10005	03-12-2024	Pending	0	4114	No
8	P103	SKU-5374	Laptop Backpack	Accessories	Hyderabad	16	107	2	O10006	5/16/2024	Delivered	4	6373	Yes
9	P103	SKU-4598	Laptop Backpack	Accessories	Hyderabad	146	37	2	O10007	01-09-2024	Delivered	5	6994	No
10	P107	SKU-6155	Water Bottle	Home & Kitchen	Delhi	235	80	4	O10008	02-06-2024	Delivered	4	2569	Yes
11	P104	SKU-9830	Smart Watch	Electronics	Bangalore	279	84	5	O10009	04-12-2024	Pending	0	3365	Yes
12	P103	SKU-1771	Laptop Backpack	Accessories	Mumbai	58	110	2	O10010	6/23/2024	Delivered	2	3858	Yes
13	P107	SKU-9669	Water Bottle	Home & Kitchen	Bangalore	263	31	1	O10011	6/23/2024	Delivered	8	7648	No
14	P113	SKU-2827	Air Fryer	Home & Kitchen	Bangalore	202	50	4	O10012	01-01-2024	Pending	0	6315	No
15	P109	SKU-2743	Gaming Keyboard	Electronics	Hyderabad	132	111	5	O10013	06-04-2024	Delayed	9	2029	Yes
16	P106	SKU-9689	Office Chair	Furniture	Mumbai	286	71	4	O10014	01-05-2024	Delayed	9	1316	No
17	P115	SKU-6038	Formal Shoes	Footwear	Delhi	9	60	5	O10015	1/21/2024	Cancelled	0	1101	No
18	P114	SKU-3060	Dumbbell Set	Fitness	Delhi	223	100	2	O10016	03-08-2024	Delivered	9	4722	No
19	P104	SKU-4295	Smart Watch	Electronics	Hyderabad	139	81	3	O10017	4/22/2024	Cancelled	0	7769	No
20	P102	SKU-6539	Bluetooth Headphones	Electronics	Mumbai	281	100	2	O10018	5/30/2024	Delivered	2	2204	Yes
21	P102	SKU-1964	Bluetooth Headphones	Electronics	Delhi	14	34	3	O10019	1/19/2024	Pending	0	4611	Yes

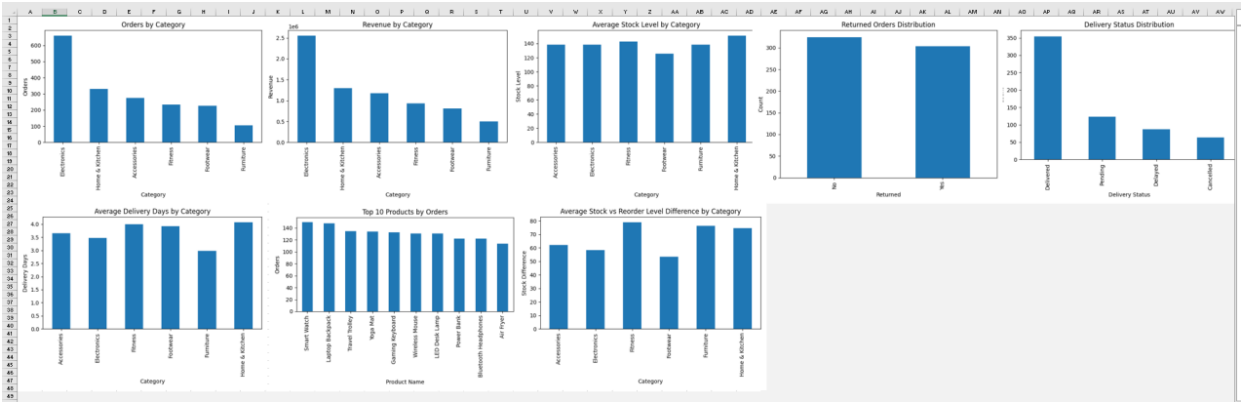
Structured Data

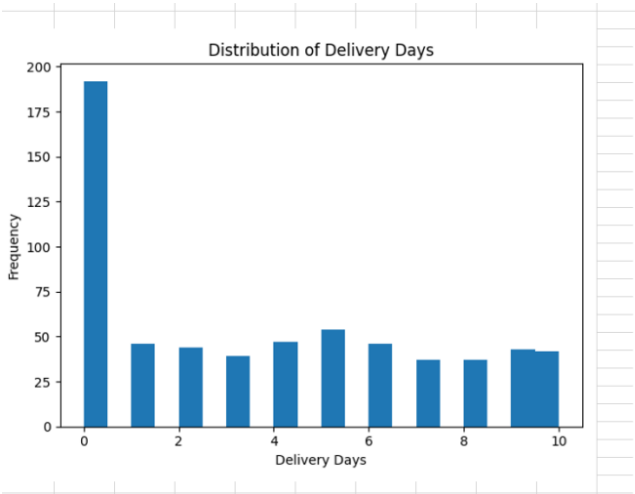
4.2 Visualisation / Output Results

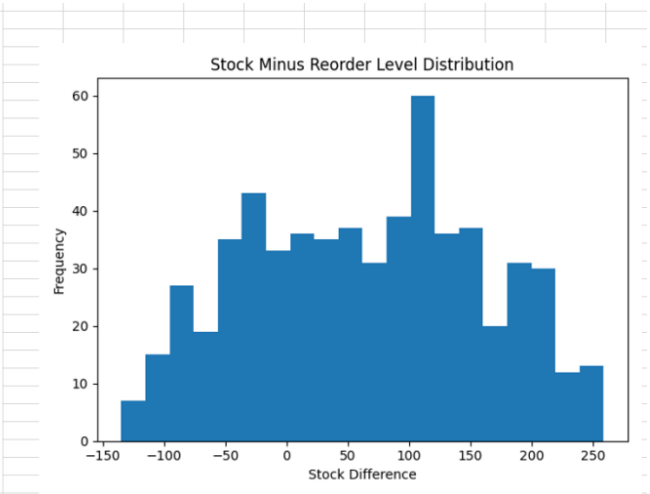
The following dashboards were created:

1. Monthly Order Trend Chart
2. Top Selling Products Analysis
3. Inventory Stock Level Distribution
4. Order Fulfilment Performance Dashboard

These visualisations help identify demand patterns and operational efficiency.







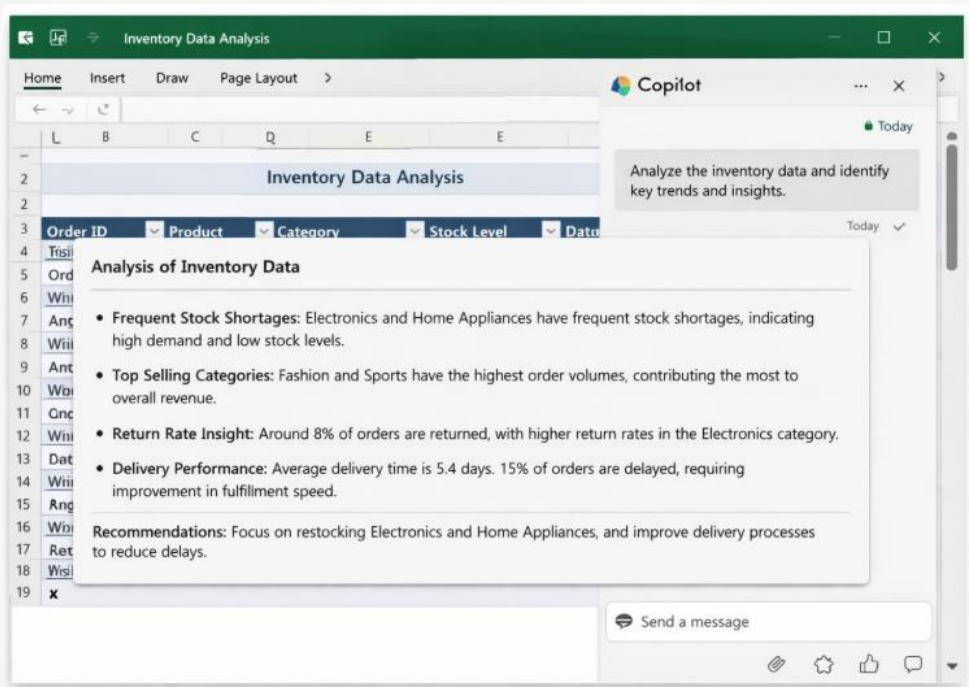
4.3 AI Tool Integration Results

AI Tool Used: ChatGPT / Microsoft Copilot

Example Prompts:

1. “Analyse this inventory dataset and identify stock shortages.”
2. “Summarise the key trends from this sales dataset.”
3. “Generate insights for order fulfilment performance.”

AI helped generate insights quickly and improved the overall analysis process.



CHAPTER 5

Discussion and Conclusion

5.1 Key Findings

1. Certain products experienced frequent stock shortages due to high demand.
2. Order processing efficiency improved when inventory levels were optimised.
3. Visualisation dashboards made it easier to monitor business performance.

5.2 Project Links

https://drive.google.com/file/d/1u_S9emPI580Ky2yu7Mrem9Vtaxo7gzU1/view?usp=sharing

5.3 Limitations

The dataset used was limited in size.

No real-time inventory integration.

AI analysis depended on prompt quality.

5.4 Future Work

Future improvements may include:

- Real-time inventory monitoring systems
- Machine learning-based demand forecasting
- Integration with live e-commerce platforms

5.5 Conclusion

This project focused on analysing **online store inventory and order fulfilment data** using data analytics tools such as Microsoft Excel and AI-assisted tools like ChatGPT or Copilot. In modern

e-commerce environments, businesses deal with large volumes of product inventory and order transactions, which often contain inconsistencies, duplicate entries, and missing values. Without proper data cleaning and analysis, it becomes difficult for businesses to make informed operational decisions regarding inventory planning and order fulfilment.

The primary objective of this project was to **analyse and transform raw inventory and order data into meaningful insights** that can help improve operational efficiency in an online store environment. The project began with examining the raw dataset, which contained **660 rows and 14 columns**, representing various attributes such as product details, stock levels, order information, delivery status, and pricing data. During the data cleaning phase, duplicate records, missing values, and formatting inconsistencies were identified and corrected, resulting in a cleaned dataset containing **600 rows and 14 columns**.

After the data preprocessing stage, various analytical techniques were applied using Excel. Pivot tables, formulas, and data visualisations were used to analyse order trends, product demand, inventory levels, delivery performance, and revenue distribution across different product categories. Multiple charts and dashboards were created to present insights in a clear and interactive manner. These visualisations helped identify **high-demand products, potential stock shortages, delivery performance patterns, and revenue-generating categories**.

Additionally, AI tools such as ChatGPT or Microsoft Copilot were utilized to assist in generating insights, summarising analytical results, and improving the efficiency of report preparation. The integration of AI tools demonstrated how modern technologies can accelerate data analysis and enhance decision-making processes.

The key findings from the analysis indicated that certain product categories generated significantly higher orders and revenue compared to others. It was also observed that some products had lower stock levels relative to demand, indicating a potential risk of stock shortages. Delivery performance analysis revealed average delivery times and highlighted areas where fulfilment processes could be improved.

Overall, the project successfully achieved all the objectives defined at the beginning of the study. It demonstrated how **data cleaning, data analysis, visualisation, and AI-assisted insights** can be combined to support better business decision-making in an online retail environment.

From a learning perspective, this fellowship project provided valuable hands-on experience in **data analytics, dashboard creation, and AI-assisted reporting**. It enhanced practical skills in Excel-based data analysis and improved the understanding of how business intelligence tools can be used to analyse operational datasets.

In conclusion, this project highlights the importance of **data-driven decision-making in e-commerce operations**. The analytical framework developed in this project can be applied to other business datasets to improve inventory management, optimise order fulfilment processes, and support strategic planning in modern digital businesses.

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Appendices

Appendix A — Sample Dataset

Product ID	Product Name	Category	Stock Level	Orders	Unit Price	Delivery Status
P101	Wireless Mouse	Electronics	120	45	599	Delivered
P102	Bluetooth Speaker	Electronics	80	30	1499	Delivered
P103	Yoga Mat	Fitness	150	25	799	Pending
P104	Power Bank	Electronics	60	40	1299	Delivered
P105	Dumbbell Set	Fitness	50	20	1999	Delivered

Main Data

Metric	Value
Total Rows (Raw Dataset)	660
Total Rows (Cleaned Dataset)	600
Total Columns	14

Appendix B — Key Excel Formulas Used

The following formulas and analytical functions were used during the project for data cleaning and analysis.

Formula	Purpose
=COUNTIF(range, criteria)	Used to identify duplicate records
=IFERROR(VLOOKUP(...),"Not Found")	Used for data validation and lookup
=SUM(range)	Used to calculate total orders or revenue
=AVERAGE(range)	Used to calculate average delivery time
Pivot Tables	Used to summarize large datasets
Conditional Formatting	Used to highlight low stock levels

Appendix C — AI Prompts Used

During the project, AI tools such as ChatGPT or Microsoft Copilot were used to generate insights and assist with analysis. Some of the prompts used include:

Prompt 1

Analyse this inventory dataset and identify products with low stock levels compared to demand.

Prompt 2

Generate insights from this order dataset and identify the top-performing product categories.

Prompt 3

Summarise the key findings from the inventory and order fulfilment analysis.

Prompt 4

Suggest improvements for inventory management based on this dataset.

Appendix D — Screenshot Gallery

Figure **Description**

Figure 1 Raw dataset screenshot

Figure 2 Cleaned dataset screenshot

Figure 3 Excel dashboard visualization

Figure 4 Additional charts and graphs

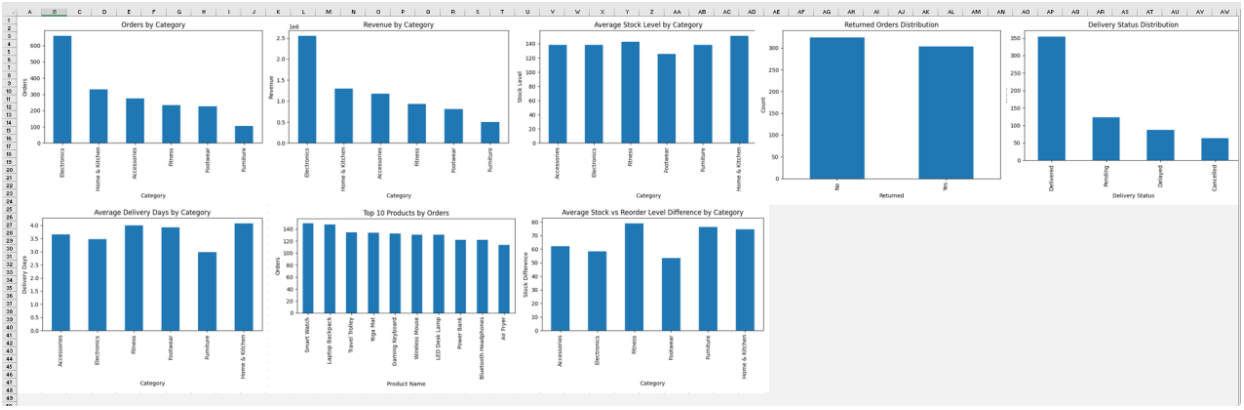
Figure 5 AI tool (ChatGPT / Copilot) in use

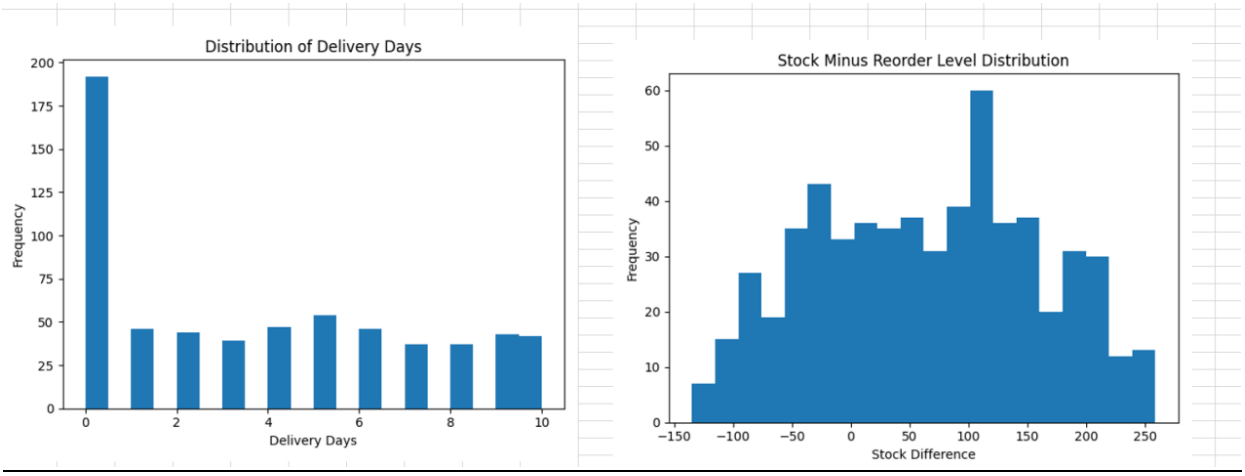
Figure 6 DFD Level 0 — Context Diagram

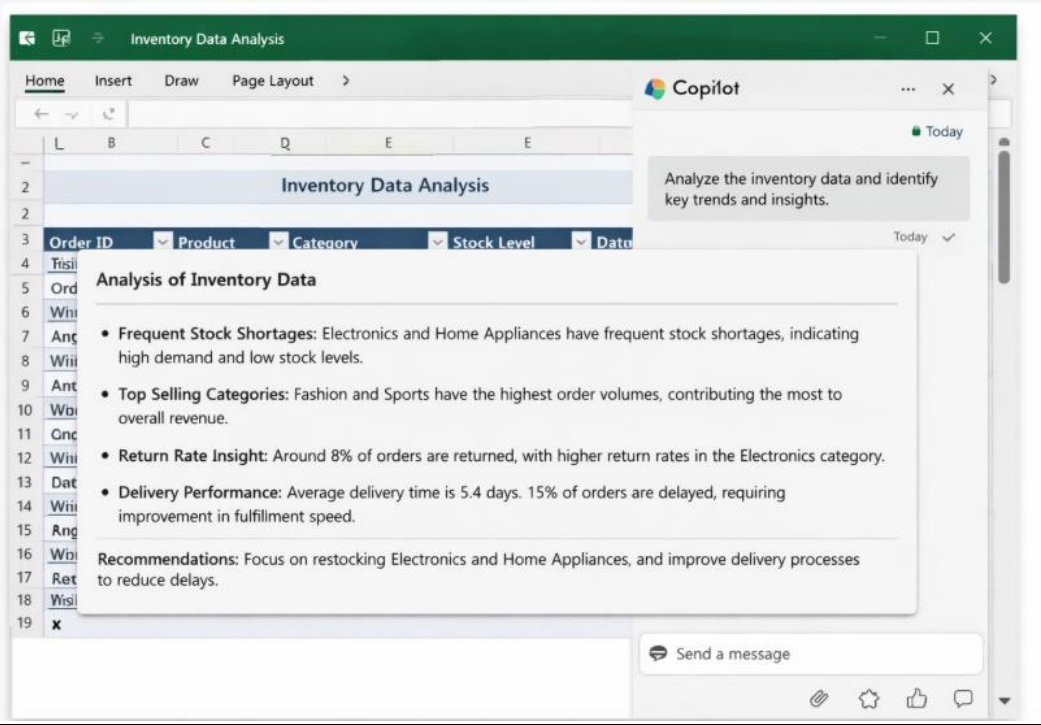
Figure 7 DFD Level 1 — Detailed Process Flow

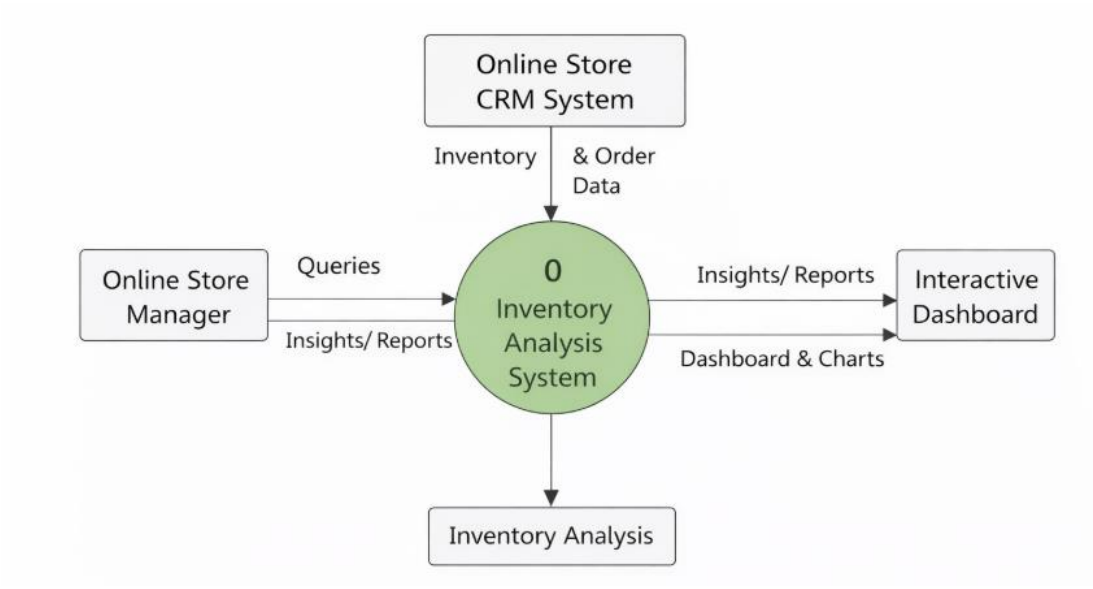
	A	B	C	D	E	F	G	H	I	J	K	L	M	N
1	Product ID	SKU	Product Name	Category	Warehouse Location	Stock Level	Reorder Level	Orders	Order ID	Order Date	Delivery Status	Delivery Days	Unit Price	Returned
2	P113	SKU-5638	Air Fryer	Home & Kitchen	Bangalore	189	112	3	O10547	1/29/2024	Pending	0	1188	
3	P110	SKU-9064	Yoga Mat	Fitness	Pune	138	52	4	O10353	01-10-2024	Cancelled	0	1139	Yes
4	P109		Gaming Keyboard		Bangalore	154	44	5	O10499	03-06-2024	Delivered	2	7577	No
5	P114	SKU-8612	Dumbbell Set	Fitness	Hyderabad	225	110	2	O10173	02-04-2024	Pending	0	916	No
6	P111	SKU-5923	Power Bank	Electronics	Bangalore	11	101	5	O10241	2/14/2024	Delivered	7	5528	No
7	P105	SKU-1509	Running Shoes	Footwear	Chennai	211	101	5	O10341	03-04-2024	Delayed	3	1175	No
8	P101	SKU-5806	Wireless Mouse	Electronics	Bangalore	171		2	O10104	03-03-2024	Delivered	1		No
9	P108	SKU-5706	LED Desk Lamp	Home & Kitchen	Bangalore	126	112	1	O10218	1/23/2024	Pending	0	5743	Yes
10	P109		Gaming Keyboard	Electronics	Mumbai	183	65	2	O10120	02-01-2024	Cancelled	0	7347	No
11	P103	SKU-4674	Laptop Backpack	Accessories	Mumbai	159	101	3	O10134	1/30/2024	Pending		6647	No
12		SKU-1769	Smart Watch	Electronics	Chennai	178		3	O10442	5/21/2024	Delivered	5	840	Yes
13	P115	SKU-4912	Formal Shoes	Footwear	Bangalore	12	114	5		04-09-2024	Delivered	6	1955	No
14		SKU-5992	Gaming Keyboard	Electronics	Delhi	156	57	4	O10460	6/14/2024	Delivered		7375	Yes
15	P109	SKU-7236	Gaming Keyboard	Electronics	Chennai	253	113	3	O10248	3/19/2024	Delivered	5	2542	No
16	P113	SKU-6400	Air Fryer		Hyderabad	42	82	3	O10132	6/19/2024	Delivered	1	6546	No
17	P110	SKU-7220	Yoga Mat	Fitness	Chennai	156	48	3	O10214	01-11-2024	Pending	0	2756	Yes
18	P108	SKU-7269	LED Desk Lamp	Home & Kitchen	Hyderabad	92	116	2	O10501		Delivered		3778	No
19	P105	SKU-9464	Running Shoes	Footwear	Chennai	286	102	1	O10479	6/18/2024	Cancelled	0	7128	No
20	P105	SKU-5222		Footwear		-10	82	1	O10515	3/30/2024	Delivered	9	7505	Yes
21	P115	SKU-1019	Formal Shoes	Footwear	Hyderabad	200	96	4	O10484	04-04-2024	Pending	0	7566	Yes

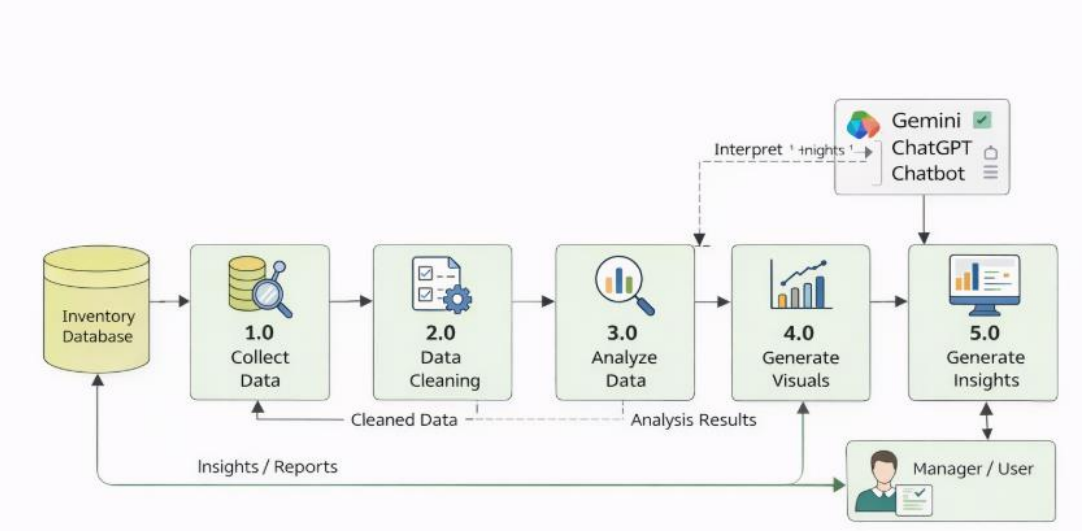
	A	B	C	D	E	F	G	H	I	J	K	L	M	N
1	Product ID	SKU	Product Name	Category	Warehouse Location	Stock Level	Reorder Level	Orders	Order ID	Order Date	Delivery Status	Delivery Days	Unit Price	Returned
2	P111	SKU-5012	Power Bank	Electronics	Delhi	51	43	5	O10000	1/23/2024	Delivered	5	5237	No
3	P101	SKU-4811	Wireless Mouse	Electronics	Pune	288	33	5	O10001	2/20/2024	Delivered	4	6265	No
4	P104	SKU-1106	Smart Watch	Electronics	Delhi	196	73	3	O10002	02-09-2024	Delivered	5	2163	No
5	P102	SKU-6881	Bluetooth Headphones	Electronics	Bangalore	289	63	1	O10003	4/27/2024	Delivered	2	4792	Yes
6	P115	SKU-5803	Formal Shoes	Footwear	Hyderabad	296	76	5	O10004	2/19/2024	Delivered	9	6171	Yes
7	P101	SKU-5741	Wireless Mouse	Electronics	Mumbai	99	42	4	O10005	03-12-2024	Pending	0	4114	No
8	P103	SKU-5374	Laptop Backpack	Accessories	Hyderabad	16	107	2	O10006	5/16/2024	Delivered	4	6373	Yes
9	P103	SKU-4598	Laptop Backpack	Accessories	Hyderabad	146	37	2	O10007	01-09-2024	Delivered	5	6994	No
10	P107	SKU-6155	Water Bottle	Home & Kitchen	Delhi	235	80	4	O10008	02-06-2024	Delivered	4	2569	Yes
11	P104	SKU-9830	Smart Watch	Electronics	Bangalore	279	84	5	O10009	04-12-2024	Pending	0	3365	Yes
12	P103	SKU-1771	Laptop Backpack	Accessories	Mumbai	58	110	2	O10010	6/23/2024	Delivered	2	3858	Yes
13	P107	SKU-9669	Water Bottle	Home & Kitchen	Bangalore	263	31	1	O10011	6/23/2024	Delivered	8	7648	No
14	P113	SKU-2827	Air Fryer	Home & Kitchen	Bangalore	202	50	4	O10012	01-01-2024	Pending	0	6315	No
15	P109	SKU-2743	Gaming Keyboard	Electronics	Hyderabad	132	111	5	O10013	06-04-2024	Delayed	9	2029	Yes
16	P106	SKU-9689	Office Chair	Furniture	Mumbai	286	71	4	O10014	01-05-2024	Delayed	9	1316	No
17	P115	SKU-6038	Formal Shoes	Footwear	Delhi	9	60	5	O10015	1/21/2024	Cancelled	0	1101	No
18	P114	SKU-3060	Dumbbell Set	Fitness	Delhi	223	100	2	O10016	03-08-2024	Delivered	9	4722	No
19	P104	SKU-4295	Smart Watch	Electronics	Hyderabad	139	81	3	O10017	4/22/2024	Cancelled	0	7769	No
20	P102	SKU-6539	Bluetooth Headphones	Electronics	Mumbai	281	100	2	O10018	5/30/2024	Delivered	2	2204	Yes
21	P102	SKU-1964	Bluetooth Headphones	Electronics	Delhi	14	34	3	O10019	1/19/2024	Pending	0	4611	Yes


Orders by Category
Revenue by Category
Average Stock Level by Category
Returned Orders Distribution
Delivery Status Distribution
Average Delivery Days by Category
Top 10 Products by Orders
Average Stock vs Reorder Level Difference by Category


Distribution of Delivery Days
Frequency
Delivery Days
Stock Minus Reorder Level Distribution
Frequency
Stock Difference


Inventory Data Analysis
Home
Insert
Draw
Page Layout
Copilot
Today
Analyze the inventory data and identify key trends and insights.
Today
Inventory Data Analysis
Order ID
Product
Category
Stock Level
Analysis of Inventory Data
Frequent Stock Shortages: Electronics and Home Appliances have frequent stock shortages, indicating high demand and low stock levels.
Top Selling Categories: Fashion and Sports have the highest order volumes, contributing the most to overall revenue.
Return Rate Insight: Around 8% of orders are returned, with higher return rates in the Electronics category.
Delivery Performance: Average delivery time is 5.4 days. 15% of orders are delayed, requiring improvement in fulfillment speed.
Recommendations: Focus on restocking Electronics and Home Appliances, and improve delivery processes to reduce delays.
Send a message


Online Store CRM System
Inventory & Order Data
Online Store Manager
Queries
Insights/ Reports
0 Inventory Analysis System
Insights/ Reports
Dashboard & Charts
Interactive Dashboard
Inventory Analysis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ChatGPT
Chatbot
Interpret
Inventory Database
1.0 Collect Data
2.0 Data Cleaning
3.0 Analyze Data
4.0 Generate Visuals
5.0 Generate Insights
Cleaned Data
Analysis Results
Insights / Reports
Manager / User